Q1.

A curve has equation $y = 4\sqrt{x}$.

(a) Show that the length of arc s of the curve between the points where $x = 0$ and $x = 1$ is given by

$$s = \int_0^1 \sqrt{\frac{x+4}{x}}\,dx$$

(4)

(b) (i) Use the substitution $x = 4\sinh^2\theta$ to show that

$$\int \sqrt{\frac{x+4}{x}}\,dx = \int 8\cosh^2\theta\,d\theta$$

(5)

(ii) Hence show that $s = 4\sinh^{-1} 0.5 + \sqrt{5}$

(6)

(Total 15 marks)